

Forward

...
 X_i ↓

Row Parallel

MatMul	$X_i \times W_{Ri} = Y_i$	} Overlap
ReduceScatter	$Y_i \rightarrow P_i$	
DimExpand	$P_i \rightarrow \bar{P}_i$	

$\bar{P}_i$ ↓

Column Parallel

DimRecover	$\bar{P}_i \rightarrow P_i$	} Overlap
AllGather	$P_i \rightarrow Y$	
MatMul	$Y \times W_{Ci} = Z_i$	

Z_i ↓
...

Backward

...
 ∇_{X_i} ↑

Row Parallel

Weight Gradient	$\nabla_{W_{Ri}} = \nabla_Y^\top W_{Ri}$
Input Gradient	$\nabla_{X_i} = \nabla_Y W_{Ri}$

∇_Y ↑

Column Parallel

AllGather	$\bar{\nabla}_Y \rightarrow \nabla_Y$	} Overlap
Weight Gradient	$\nabla_{W_{Ci}} = \nabla_{Z_i}^\top Y$	
ReduceScatter	$\nabla_{Y_i} \rightarrow \bar{\nabla}_Y$	} Overlap
Input Gradient	$\nabla_{Y_i} = \nabla_{Z_i} W_{Ci}$	

∇_{Z_i} ↑
...